

ROSHAN SRINIVASAN

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Portfolio — roshansrini26 — roshansrini26

SUMMARY

Data Science master's student with hands-on experience in data analytics, machine learning, and applied projects. Strong analytical skills, experience in Agile teams, and a fast learner of new tools and technologies with a practical, solution-oriented mindset.

EDUCATION

Technical University of Hamburg-Harburg

Master of Science

Specialization: Data Science

10/2025 – Present

Hamburg, Germany

Amrita University

Bachelors in Computer Science Engineering (8.44)

2021 – 2025

Chennai, India

TECHNICAL EXPERIENCE

Formula Student Team - Egnition

Web Developer & Content Writer

11/2025 – Present

Hamburg, Germany

- Maintaining and optimizing Formula Student team website, improving sponsor visibility and professional presentation.
- Creating marketing and promotional assets aligned with sponsor branding for events, workshops, and public demonstrations.
- Assisting in planning and execution of workshops and showcases.

Motiv AI LLP

Data Science and Analytics Intern

12/2024 – 04/2025

Coimbatore, India

- Developed ML-based Transport Mode and Crash Detection systems used in over 100 pilot deployments, reducing false alarms by 75% and enabling data-driven safety analytics for all the trips.
- Improved data reliability by standardizing collection workflows, increasing usable data share from 25% to 40% and enabling smooth integration with IIT Madras (CoEZET) research pipelines.
- Applied algorithmic optimization and data-driven decision making to enhance model performance and efficiency for real-world deployment.

PROJECTS

Digital Scholarship Platform — Walwin | Python, NLP, PostgreSQL, Docker, Agile (SCRUM)

11/2025 – Present

- Conceptualized a scholarship recommendation system aligning applicant profiles with eligibility criteria to improve matching efficiency.
- Served as SCRUM Master, leading Agile ceremonies (sprint planning, stand-ups, reviews), coordinating cross-functional collaboration, and ensuring timely progress.
- Project conducted under TUHH's "Fishing for Experience" program, leading to an Agile-certified project experience.

Quantum-Assisted Deep Learning for Secure Brain Tumour Image Classification –

04/2025

Final year project

Quantum Circuits, Deep Neural Networks, Cryptography, Python

- Designed quantum circuits for image preprocessing, cutting per-image computation time by approximately 70% compared to the classical baseline. Integrated cryptographic techniques to ensure secure image transmission, preserving data integrity and patient privacy.
- Implemented and optimized a Deep Neural Network (DNN) to improve classification accuracy while reducing training time. Executed comparative analysis between quantum-assisted and classical deep learning approaches resulting 75% reduced running time, demonstrating efficiency gains.

Fitness Tracker for Barbell Exercises | Python, NumPy, Pandas, Scikit-learn, TensorFlow, Matplotlib

01/2025 – 02/2025

- Created an end-to-end machine learning pipeline to classify six barbell exercises using raw sensor data, improving data quality by applying advanced outlier detection techniques, including Chauvenet’s Criterion, to ensure robust model inputs.
- Performed feature extraction and dimensionality reduction using PCA and Fourier Transformation to capture patterns, then trained and evaluated multiple models (Neural Networks, Random Forest, Decision Tree), achieving 99% classification accuracy with the Neural Network.
- Demonstrated strong capability in signal processing, feature engineering, and model optimization.

TECHNICAL SKILLS

Programming Languages: Python, C++
Frameworks & Libraries: TensorFlow, OpenCV, Hugging Face, BERT
Web Technologies: HTML5, CSS3, Bootstrap, Angular
Databases: MySQL, MongoDB
Tools: Jupyter (JupyterLab, Jupyter Notebook), Git/GitHub, Microsoft Power BI, MySQL Workbench
Specialized Domains: Deep Learning, Computer Vision, Natural Language Processing, Cloud Computing (AWS)

RESEARCH PAPER

Optimizing Lung Cancer Classification Prediction based on Random Forest and SMOTE Techniques (2024)

- Developed a Random Forest-based classification model for lung cancer prediction, achieving 98% accuracy and 99% F1-score.
- Addressed severe class imbalance using SMOTE, significantly improving model robustness and predictive performance.
- Showcased effective use of ensemble learning and data-level optimization for healthcare analytics.

CERTIFICATES

AWS Certified Cloud Practitioner 02/2025 – 02/2028

Introduction to Big Data with Spark and Hadoop – IBM

Database and SQL for Data Science with Python – IBM

NETWORKS AND LEADERSHIP

Google Developers School Club 01/2024 – 03/2024
National-Level Techfest Organizer – Tantrotsav’24 Chennai, India

- Led the coordination of a two-day national techfest, managing logistics, securing sponsorships, and ensuring the smooth execution of technical events and activities.

Google Developers Student Club AVV Chennai 08/2023 – 07/2024
Management Lead Chennai, India

- Organized workshops and events in collaboration with sponsors and Google industry experts on Google Cloud Platform and machine learning topics.

AMC FOSS Club 01/2023 – 05/2023
Android Development Mentor Chennai, India

- Guided students in Android development using Android Studio, supporting them in building practical projects.
- Provided hands-on assistance with UI design, database integration, and app deployment to help students deliver functional applications.

LANGUAGE SKILLS

English: Proficiency Level C1
German: Proficiency Level A2